

# CSE2315 Lab Course

## Solutions 2

**deadline:** February 19, 2026, 13:45

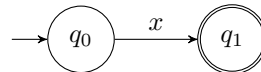
1. Consider the following language:

$$L = (a^*b)^*.$$

Construct a NFA  $N$  for which  $L(N) = L$ . To this end, please answer the following questions.

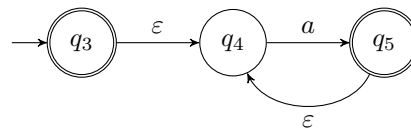
- (a) Consider a symbol  $x$ . Construct a NFA  $N_1$  for which  $L(N_1) = \{x\}$ .

**Solution:** The following NFA has the language  $\{x\}$ .

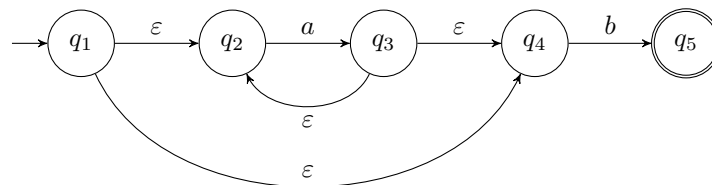


- (b) Construct the NFA  $N$  so that  $L(N) = L$  in a systematic way. Use your construction in (a) as well as the constructions used in the proofs of Theorem 1.47 and 1.49 in Sipser. Please give the resulting NFA and do not simplify it.

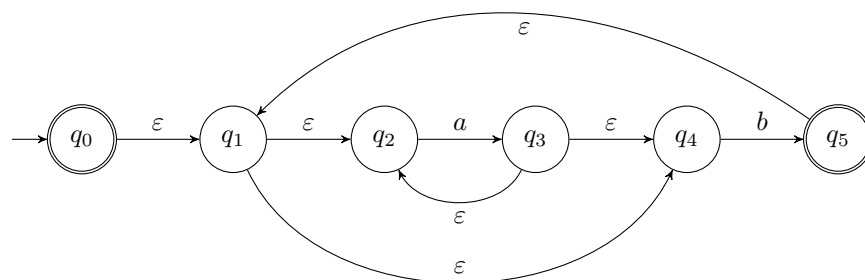
**Solution:** We show each step of building the NFA bottom-up. (This was not necessary.) First, to get  $a^*$ , use (a) and Figure 1.50:



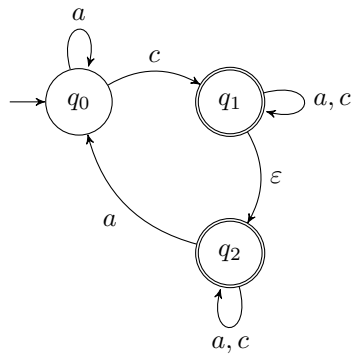
Now to get  $a^*b$ , use (a) and Figure 1.48:



Finally, to get  $(a^*b)^*$ , apply Figure 1.50 once more to get the following NFA. This NFA indeed has language  $L$ .

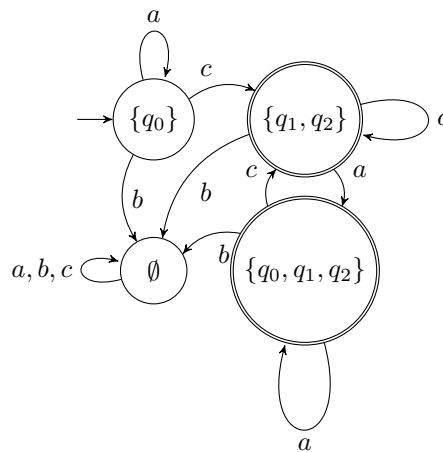


2. Taken from Resit 2021-2022: Consider the NFA  $N = (\{q_0, q_1, q_2\}, \{a, b, c\}, \delta, q_0, \{q_1, q_2\})$  where  $\delta$  is depicted using the transition diagram below.

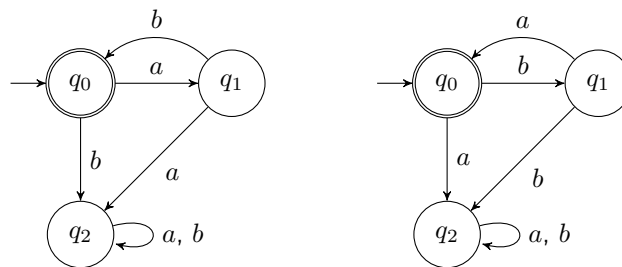


Construct a DFA  $D$  such that  $L(D) = L(N)$ . Use the procedure given in Sipser. Please give the resulting DFA and leave out unreachable states.

**Solution:**



3. Consider the following two DFAs  $D_1, D_2$ :



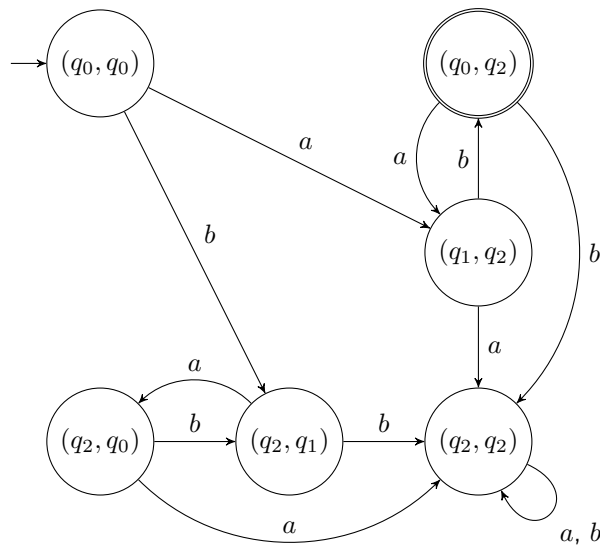
(a) Describe the languages of  $D_1$  and  $D_2$ .

**Solution:**

The language of  $D_1$  is  $L(D_1) = \{(ab)^n \mid n \geq 0\} = \varepsilon, ab, abab, \dots$

The language of  $D_2$  is  $L(D_2) = \{(ba)^n \mid n \geq 0\} = \varepsilon, ba, baba, \dots$

(b) Construct a DFA  $D$  such that  $L(D) = L(D_1) \cap L(D_2)^c$ , where  $L(D_2)^c$  is the complement of the language of  $D_2$ . Adapt the procedure given in the proof of Theorem 1.25 in Sipser. Leave out unreachable states.

**Solution:**

4. Suppose we have the following language  $L$  over the alphabet  $\Sigma = \{a, b, c\}$ :

$$L = \{w \in \Sigma^* \mid w \text{ has an } a \text{ and every } a \text{ after the first } a \text{ in } w \text{ is immediately followed by a } c\}.$$

For example  $aacb$  is in the language, so is  $acbac$  (note that it is okay for the first  $a$  to also be followed by a  $c$ ). Words like  $abca$  and  $abaab$  are not in the language.

- (a) Give a regular expression  $R$  such that  $L(R) = L$ .

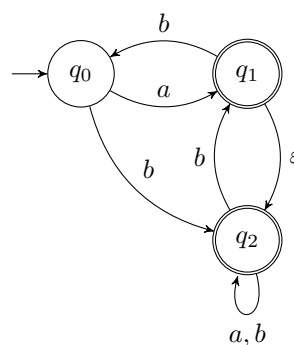
**Solution:**

$$R = (b \cup c)^* a (ac \cup b \cup c)^*$$

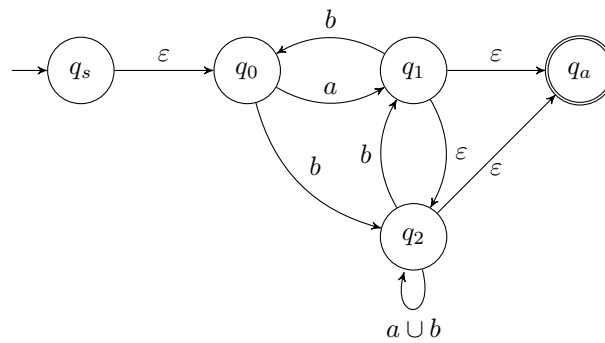
- (b) Explain how you arrived at your answer.

**Solution:** First we allow any character other than  $a$ , then we read our first  $a$  and after that  $as$  can only be followed by  $cs$ .

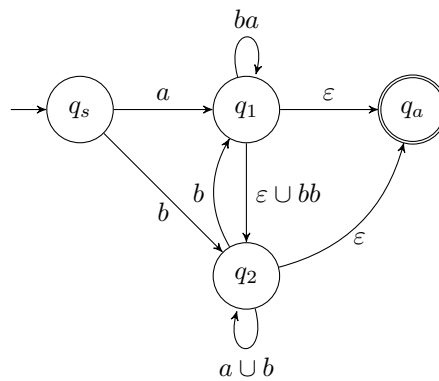
5. Convert the following NFA  $N$  to a regular expression  $R$  such that  $L(R) = L(N)$ . Please show all the intermediate steps and eliminate the states in the following order:  $q_0, q_1, q_2$ .



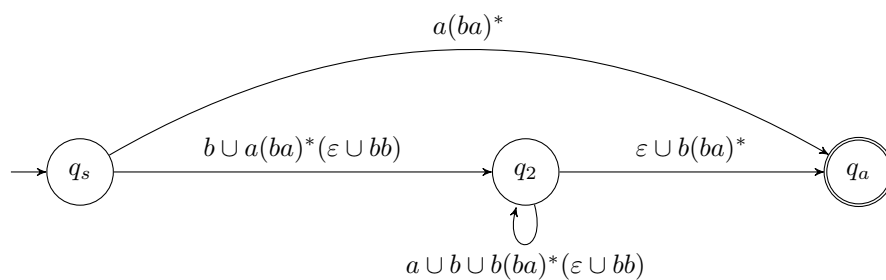
**Solution:** Rewrite to a GNFA by adding a new start state  $q_s$  and accept state  $q_a$  and turning the labels into regular expressions:



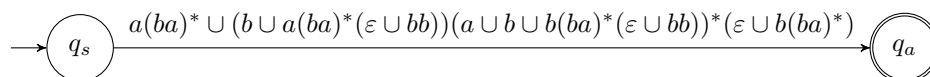
Eliminate  $q_0$ :



Eliminate  $q_1$ :

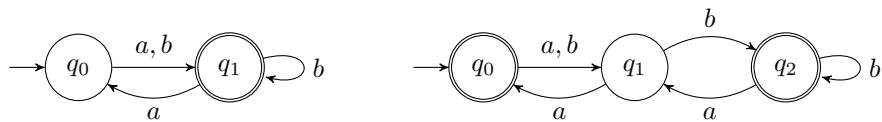


Eliminate  $q_2$ :



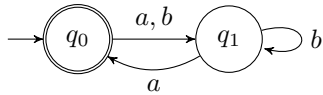
The desired regular expression can be seen on the last remaining transition.

6. *Revision, old exam question midterm 18-19* Consider the following two DFAs,  $D$  on the left and  $D'$  on the right:

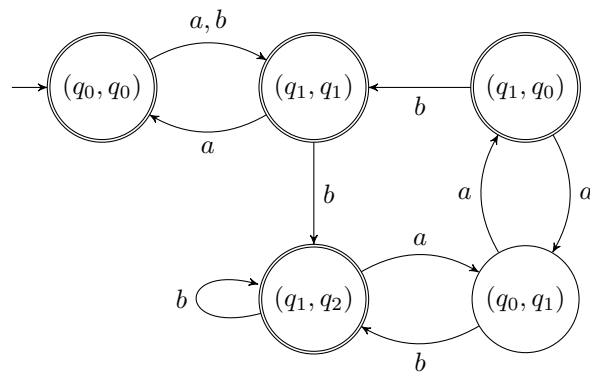


- (a) Construct a new DFA  $D''$  of at most 2 states, such that  $L(D'') = \overline{L(D)}$ . A transition diagram and a short explanation suffice.

**Solution:** We should simply invert the accepting states.



- (b) Use the procedure from Sipser to construct a DFA  $D'''$ , such that  $L(D''') = L(D) \cup L(D')$ . You should leave out unreachable states.



**Solution:**

See next page for an optional extra exercise puzzle.

**Optional extra exercise** This last exercise is not part of the learning material. However, if you like puzzles and have some time left, you might like to do this challenging puzzle.

Although your adventure of last week was a little more dangerous than expected, you decide it was worth the treasure and even want to find some more. However, this time you should be a bit better prepared. Searching for information, you find a bit about how the teleportation devices work. They deconstruct you to atoms at one place, send the information of how that was done to another place and construct you back to normal there, with completely new atoms. But since you are now consisting of different matter, are you still you? You decide to not ponder about that too much and search further. You find another mystical mansion somewhere but you read that it is a bit different than the previous one. While the teleportation devices from the other mansion always build you back up in exactly 1 place, the teleporation devices from this mansion might build you up in 0 or multiple places. Multiple places might not be so bad (or maybe it is, depending on how much you like yourself), but if you are build up in 0 places you just dissappear (how dare they still call that a teleportation device!). Because of this danger, you decide to build a remote-controlled robot. On the control are 2 buttons, one to let the robot take the left teleportation device, one to let it take the right one. Whenever the robot enters a room, you get a message on a screen telling you what colour the walls are.

You send in the robot using a teleportation device and let it take the treasure (with a different button I forgot to mention earlier). The walls of the first room are grey. Then, you press the button that lets the robot go left. This time, you get 2 messages: 1 tells you that the robot is in a grey room, the other that the robot is in a white room. You press a few more buttons (given below), but then the walls start to crumble. This mansion is falling apart as well. There is only time to use 6 teleportation devices. How can you be sure to get the robot with the treasure out?

| Start  | Left         | Right   | Left   | Right        | Left         | Left         | Left         |
|--------|--------------|---------|--------|--------------|--------------|--------------|--------------|
| {Grey} | {Grey,White} | {White} | {Grey} | {Grey,White} | {Grey,White} | {Grey,White} | {Grey,White} |

| Left              | Right        | Left         | Left         | Left         | Right   | Left   |
|-------------------|--------------|--------------|--------------|--------------|---------|--------|
| {Grey,Grey,White} | {Grey,White} | {Grey,White} | {Grey,White} | {Grey,White} | {White} | {Grey} |

- Each room has 2 teleportation devices, a left one and a right one
- Teleportation devices are deterministic (the same device in the same room will always lead to the same room(s))
- If 2 (or more) teleportation devices teleport the same thing to the same room at the same time, only 1 of those will be build (the device cannot see that they are 2 signals)
- At least 1 teleportation device leads outside
- There are at most 4 rooms
- Rooms will not change colour (the same room will always have the same colour)
- After the robot has taken the teleportation devices as given in the table (and entered rooms with wall colours as given in the table), how can you get the robot outside to collect the treasure?

**Solution:** *Left → Right*

- Extra: How can you get 2 treasures?

**Solution:** *Right → Left → Right → Left → Right*

- Extra: How can you get 3 treasures? (here, you have to think outside the box a little)

**Solution:** While letting the robot move  $Right \rightarrow Left \rightarrow Right \rightarrow Left$ , teleport inside the mansion yourself and move  $Left \rightarrow Left$ . When you are both done, there should be a robot in your room. Pick that up and take the left teleportation device with it (if needed, you can press  $Right$  to clear the way in the final room). Then move together with the robot to the right teleportation device (either by pushing  $Right$  or by just carrying it) to leave the mansion. Note that this will construct some copies of you who will either have to step in a teleportation device that deconstructs them or be stranded in the mansion while it's falling apart, so this might not be desired depending on your ethical opinion on this.

- Extra: Given that you have time left to use  $n$  teleporation devices (instead of 6), how many treasures would you be able to collect?

**Solution:**  $Left \rightarrow Right$  will get you 1 treasure. Alternating  $Right$  and  $Left$  (starting at  $Right$ ), will let you get  $(n-1)/2$  treasures. Constantly pressing  $Left$ , while going in yourself and pushing the robot in the last room in the right teleportation device will let you get  $n-5$  treasures. So the maximum treasures you could get is:

$$amount(n) = \begin{cases} 0 & \text{for } n \leq 1 \\ 1 & \text{for } 2 \leq n \leq 4 \\ \lfloor (n-1)/2 \rfloor & \text{for } 5 \leq n \leq 9 \\ n-5 & \text{for } 10 \leq n \end{cases}$$